

Week 2 Project 1: Sale Performance Analysis system.

Dataset

- data
- Category
- Product
- Quantity
- sales

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("product_sales_dataset_final.csv")
```

```
df.head()
```

	Order_ID	Order_Date	Customer_Name	City	State	Region	Country	Category	Sub_Category	Product_Name	Quantity
0	1.0	08-23-23	Bianca Brown	Jackson	Mississippi	South	United States	Accessories	Small Electronics	Phone Case	
1	2.0	12-20-24	Jared Edwards	Grand Rapids	Michigan	Centre	United States	Accessories	Small Electronics	Charging Cable	
2	3.0	01-29-24	Susan Valdez	Minneapolis	Minnesota	Centre	United States	Clothing & Apparel	Sportswear	Nike Air Force 1	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49999 entries, 0 to 49998
Data columns (total 14 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Order_ID        10000 non-null  float64
1   Order_Date      10000 non-null  object
2   Customer_Name   10000 non-null  object
3   City            10000 non-null  object
4   State           10000 non-null  object
5   Region          10000 non-null  object
6   Country         10000 non-null  object
7   Category        10000 non-null  object
8   Sub_Category    10000 non-null  object
9   Product_Name    10000 non-null  object
10  Quantity        10000 non-null  float64
11  Unit_Price      10000 non-null  float64
12  Revenue         10000 non-null  float64
13  Profit          10000 non-null  float64
dtypes: float64(5), object(9)
memory usage: 5.3+ MB
```

```
df.describe()
```

	Order_ID	Quantity	Unit_Price	Revenue	Profit
count	10000.00000	10000.000000	10000.000000	10000.000000	10000.000000
mean	5000.50000	1.849600	385.353689	715.869528	157.122257
std	2886.89568	1.088714	277.509694	738.705012	150.634933
min	1.00000	1.000000	19.730000	20.300000	4.360000
25%	2500.75000	1.000000	164.827500	230.222500	59.320000
50%	5000.50000	1.000000	308.140000	473.755000	110.405000
75%	7500.25000	2.000000	566.147500	890.640000	203.160000
max	10000.00000	8.000000	1372.890000	7110.960000	1760.890000

```
df.isnull().sum()
```

```

      0
Order_ID      39999
Order_Date      39999
Customer_Name  39999
City           39999
State          39999
Region         39999
Country        39999
Category       39999
Sub_Category   39999
Product_Name   39999
Quantity       39999
Unit_Price     39999
Revenue        39999
Profit         39999

```

```
dtype: int64
```

```
df.fillna(0, inplace=True)
```

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], errors='coerce')
```

```
print(df.columns)
```

```

Index(['Order_ID', 'Order_Date', 'Customer_Name', 'City', 'State', 'Region',
      'Country', 'Category', 'Sub_Category', 'Product_Name', 'Quantity',
      'Unit_Price', 'Revenue', 'Profit'],
      dtype='object')

```

```

total_sales = df['Revenue'].sum()
print("Total Revenue:", total_sales)

```

```
Total Revenue: 7158695.279999999
```

```

if 'Sales' in df.columns:
    total_sales = df['Sales'].sum()
    print("Total Sales:", total_sales)
else:
    print("Sales column not found")

```

```
Sales column not found
```

```

average_sales = df['Revenue'].mean()
print("Average Sales:", average_sales)

```

```
Average Sales: 143.1767691353827
```

```

category_sales = df.groupby('Category')['Revenue'].sum().sort_values(ascending=False)
print(category_sales)

```

```

Category
Electronics      2923782.77
Home & Furniture  2385671.95
Clothing & Apparel 1350706.07
Accessories      498534.49
0                0.00
Name: Revenue, dtype: float64

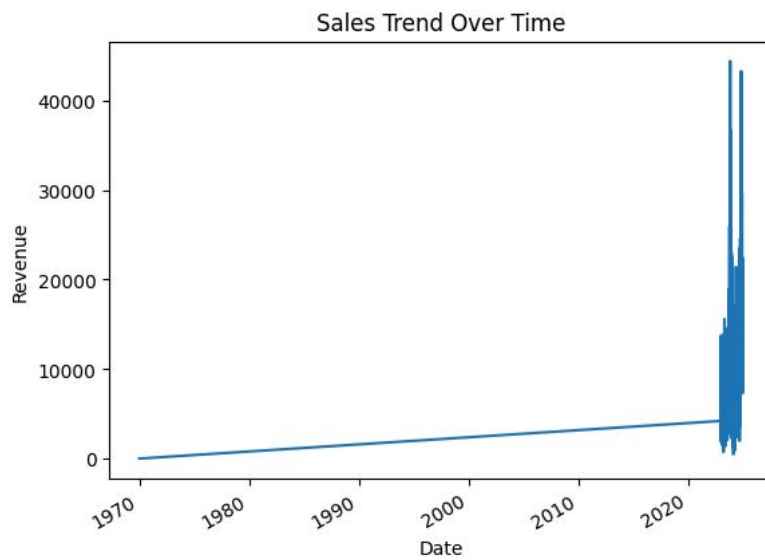
```

```
daily_sales = df.groupby('Order_Date')['Revenue'].sum()
```

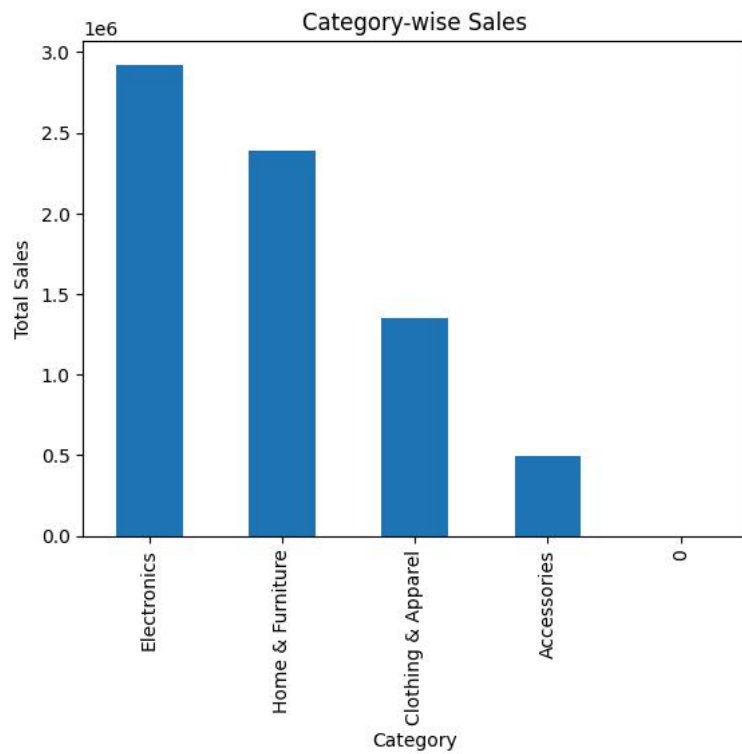
```

plt.figure()
daily_sales.plot()
plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Revenue")
plt.show()

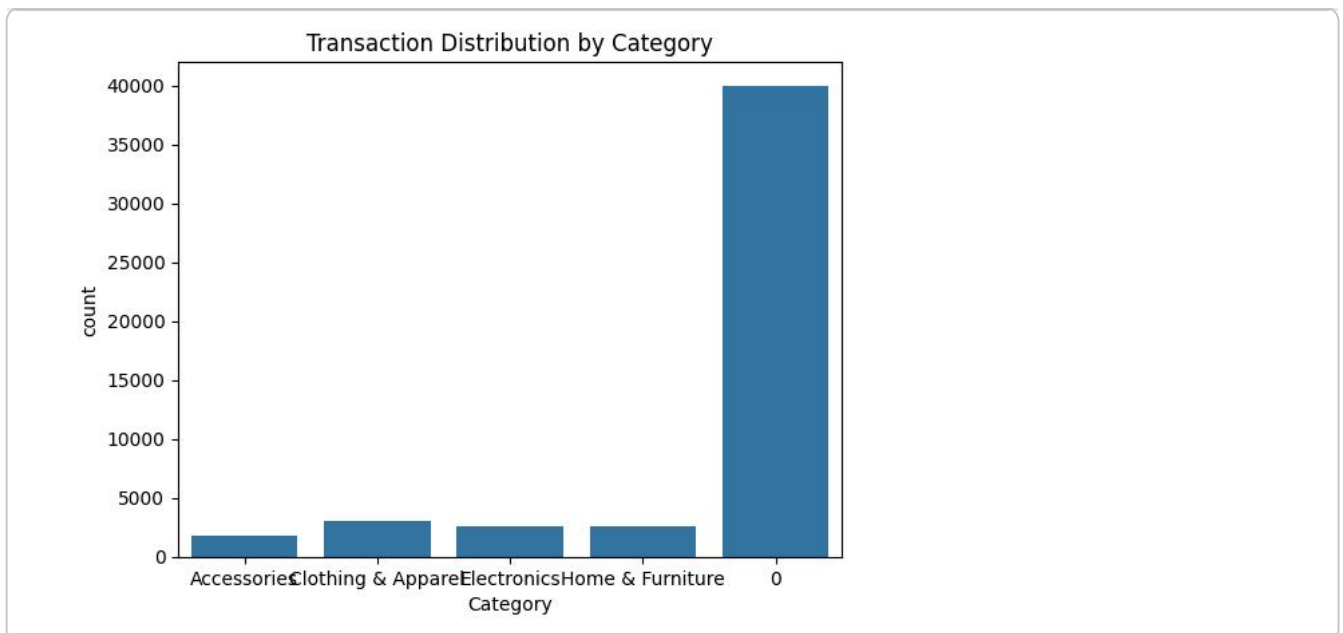
```



```
plt.figure()
category_sales.plot(kind='bar')
plt.title("Category-wise Sales")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.show()
```



```
plt.figure()
sns.countplot(x='Category', data=df)
plt.title("Transaction Distribution by Category")
plt.show()
```



This sales Performance Analysis System successfully demonstrates how Python and Pandas can be used to clean, analyze, and visualize sales data.

The insights gained help organizations understand performance trends and make data-driven decisions.

- Read the sales dataset.
- Clean and check the data.
- Analyze sales performance.
- Find trends and category-wise sales.
- Show results using graphs.

Dataset Name: product_sales_datast_final.csv

File type: CSV file.

- Sales data is very important for any business.
- It helps organizations understand how much they are selling, Which products are performing well, and where improvements are needed.
- sales data is usually large and difficult to understand in raw form.